Copyright and License Agreement History Management Framework for Outsourced Software

Taeyang Lee
Dept. of Computer and Information Security & Convergence Engineering for Intelligent Drone
Sejong University
Seoul, Republic of Korea
taeyang@pel.sejong.ac.kr

Jinsue Lee
Dept. of Computer and Information Security & Convergence Engineering for Intelligent Drone
Sejong University
Seoul, Republic of Korea
jinsue@pel.sejong.ac.kr

Seungchan Woo
Dept. of Computer and Information Security
Sejong University
Seoul, Republic of Korea
seungchan@pel.sejong.ac.kr

Jong-Hyouk Lee
Dept. of Computer and Information Security & Convergence Engineering for Intelligent Drone
Sejong University
Seoul, Republic of Korea
jonghyouk@sejong.ac.kr

***Abstract*—The practice of transitioning software development tasks to outsourcing has become increasingly prevalent. However, this approach presents challenges in systematically managing the historical records of copyright and license agreements associated with outsourced software. To address potential conflicts between clients and vendors in outsourced software transactions, this paper introduces a framework for effectively managing the history of copyright and license agreements pertaining to outsourced software.**

***Index Terms*—Outsourcing, SBOM, Copyright, License, Hyperledger Fabric, Smart Contract**

I. Introduction

The transition of software development tasks to outsourcing has become a common practice in the industry. However, software copyright infringement have arisen, including clients making unjust demands for access to the software's source code developed by vendors [1]. Furthermore, traditional copyright and license agreements method for outsourced software not only complicates the management of contract history but also hinders data transparency by relying on centralized systems. If the traditional method is used in the metaverse, where digital contracts are essential, it can cause resemblant problems with the real world. This paper presents a blockchain-based framework designed to address these problems.

The structure of this paper is as follows: Section II delves into the Software Bill of Materials and Smart Contracts, while Section III proposes a framework for managing the copyright and license history of outsourced software. Finally, Section IV offers a conclusion for the paper.

II. Software Bill of Materials

The Software Bill of Materials (SBOM) enumerates all components comprising a software product, along with information on dependencies and hierarchical relationships among them. The Software Package Data Exchange (SPDX) format, a widely adopted SBOM standard, is specifically designed to represent software components and their associated licensing information. By employing the SPDX format, SBOM facilitates supply chain security, transparency, and trust in software through the clear specification of copyright and license information [2].

III. Proposed Framework

While the copyright and license scope of outsourced software is documented using SBOM with SPDX each time stakeholders establish a new agreement pertaining to the software's copyright and license, generating a new SBOM containing the updated agreement details is necessary. However, no existing systems facilitate the management of agreement history through SBOMs. To address this gap, this paper introduces a blockchain-based framework that leverages the chaincode of Hyperledger Fabric to support digital agreements between outsourced software clients and vendors. The chaincode, smart contract, is a self-executing agreement represented as program code within a Hyperledger Fabric. Additionally, it employs SBOMs formatted by SPDX to manage the history of copyright and license agreements. The proposed framework ensures the transparency of agreement content by streamlining the management of outsourced software copyright and licensing agreement history.

In the proposed framework, there are major peers: agent, clients, and vendors. The agent serves as a service operator that hosts services dedicated to managing copyright and license agreements for outsourced software. Acting as a peer within the blockchain network, the agent is responsible for managing the blockchain itself, generating and enrolling chaincodes. The clients and vendors, as service consumers, can access agreement services provided by the agent. Software developers or enterprises, in their capacity as vendors, possess rights to the outsourced software by recording the relevant copyright and license information using chaincode obtained from the

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agent. This ensures that the vendor's intellectual property is protected and recognized within the blockchain network.

On the other hand, clients acquire rights to the outsourced software by agreeing to the specified terms and conditions and making payments for the software's copyright and license. This process formalizes the relationship between clients and vendors, ensuring that both parties' rights and obligations are clearly defined and upheld.

By incorporating this comprehensive framework, the management of outsourced software copyright and licensing agreements is streamlined, enabling a more transparent and efficient system for all stakeholders involved.

We have established a procedure for handling these agreements while generating SBOMs using the SPDX format. Fig. 1 illustrates the outsourced software copyright and license agreement process based on the Hyperledger Fabric, as well as the generation of SBOMs. Within the proposed procedure, several chaincodes are employed, as outlined in Table I and further defined throughout this paper.

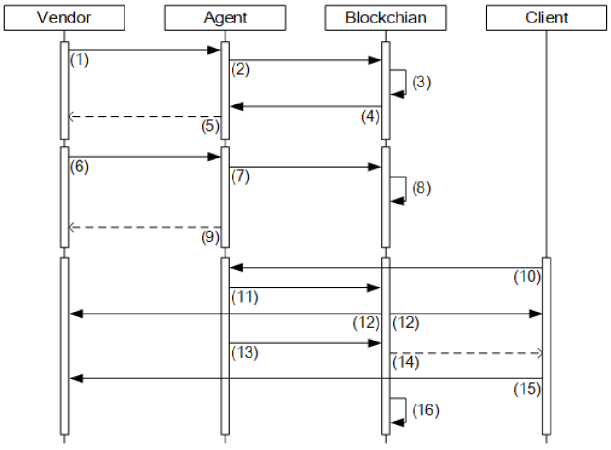


Fig. 1. Proposed process for the outsourced software copyright and license agreement

(1) Vendors initiate a request for the agent to upload the outsourced software onto the blockchain. (2) The agent deploys chaincodes: ***swUpload()*** and ***sbomGenerator()***. (3) Utilizing these chaincodes, the outsourced software is uploaded to the blockchain and an SBOM for the software is generated. (4) The agent retrieves the P2P reference hash values of the uploaded software and the SBOM from the blockchain. (5) Subsequently, the agent responds to the vendors.

(6) Vendors request the agent to upload information regarding the copyright and license of the uploaded software. (7) The agent deploys a chaincode: ***rightLicenseInfoRegister()***. (8) Using this chaincode, the SBOM, which includes the composition details of the uploaded software, is updated with the copyright and license information. This data encompasses copyright transfers, payments, and fees for software usage within the license scope. The SBOM can be located using the uploaded software ID. (9) The agent then responds to the vendors.

TABLE I
CHAINCODES FOR THE COPYRIGHT AND LICENSE AGREEMENTS

Chaincode	Description
swUpload()	Automatize the software enrollment
sbomGenerator()	Automatize the SBOMs generation
rightLicenseInfoRegister()	Automatize the copyright and license information enrollment
agreement()	Automatize agreements between clients and vendors
swDeliver()	Automatize the software delivery to clients

(10) Clients request the agent to obtain the outsourced software developed by the vendors. (11) To facilitate a copyright and license agreement between clients and vendors, the agent deploys a chaincode: ***agreement()***. (12) With ***agreement()***, clients and vendors receive the outsourced software's copyright and license information from the blockchain and negotiate terms. Once they reach a consensus, they finalize the agreement by signing the contract. (13) Following this, the agent deploys a chaincode: ***swDeliver()***. (14) Through this chaincode, the client can access the outsourced software from the blockchain. (15) Upon receipt of the software, clients pay vendors according to the payment amount specified in the contract. (16) Finally, all transaction information resulting from the previous steps is recorded on the blockchain. If clients and vendors need to modify the copyright and license agreement at a later stage, they can use ***agreement()***. In subsequent agreements, the agent retrieves the SBOM of the outsourced software from the blockchain using the software ID and SBOM ID. New copyright and license agreements are recorded on the retrieved SBOM. Blockchain nodes can query the copyright and license agreement history by searching for the SBOM with the outsourced software ID and SBOM ID.

IV. CONCLUSION

The traditional approach to managing outsourced software copyright and license agreements often struggles to ensure data transparency and maintain a comprehensive agreement history due to its reliance on centralized systems. Consequently, we have proposed the blockchain-based framework for managing the history of copyright and license agreements for outsourced software. The proposed framework enables the automatic generation of updated SBOMs whenever an agreement is altered. The suggested framework enables metaverse operators to offer digital contract services for outsourced software copyright and license agreements within the metaverse.

ACKNOWLEDGMENT

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